2025-10-16 - Valentini (5)

Previous lecture: 2025-10-10 - Valentini (4)
Next lecture: 2025-10-17 - Valentini (6)

Linearized VP equations

$$\begin{cases} \dfrac{\partial f_{\alpha 1}}{\partial t} + \bar{v}\cdot\nabla f_{\alpha 1} - \dfrac{\rho_\alpha}{m_\alpha}\nabla\phi_1\cdot\nabla_{\bar{v}} f_{\alpha 0} = 0 \\ -\nabla^2\phi_1 = 4\pi\displaystyle\sum_\alpha \rho_\alpha \int_{-\infty}^{\infty} f_{\alpha 1}\left(\bar{r},\bar{v},t\right) d^3v \end{cases}$$

VM distribution

$$f_{\alpha 0} = \frac{n_{\alpha 0} m_\alpha^{3/2}}{\left(2\pi k_B T_{\alpha 0}\right)^{3/2}} \exp\left[-\frac{m_\alpha v^2}{2k_B T_{\alpha 0}}\right]$$

$$\phi_0 = 0, \qquad n_{i0} = n_{e0} = n_0$$

Laplace transform

$$\mathscr{L}\left[f(t)\right](p) = \tilde{f}(p) = \int_0^\infty f(t)e^{-pt}\,dt$$

complex munber p_1

$$\exists p_1 \in \mathbb{C} \text{ s.t. } \int_0^\infty \left|f(t)e^{-pt}\right| dt < \infty$$

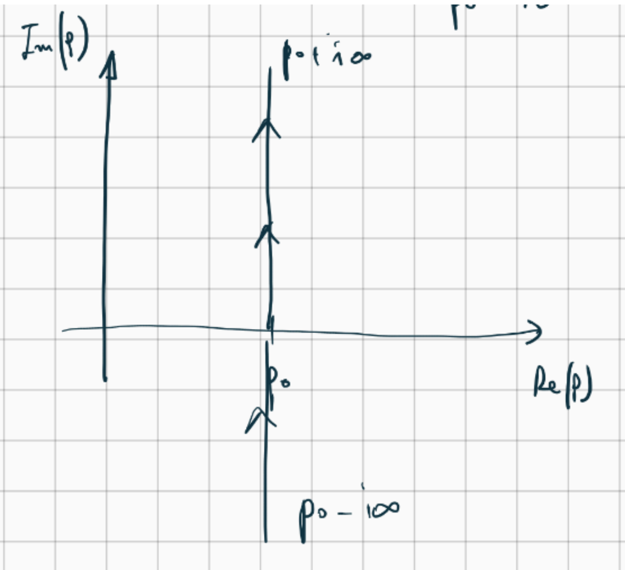
is convergent.
If there exists two constants M and $p_0 \in \mathbb{R}$ such that

$$|f(t)| \leq Me^{p_0 t} \implies \tilde{f}(p) \text{ is well defined } \forall\, \mathrm{Re}(p) \geq p_0$$

with p_0 "indice di crescita" or growth index
The inverse transform is

$$\mathscr{L}[\tilde{f}(p)]^{-1}(t) = f(t) = \frac{1}{2\pi i}\int_{p_0 - i\infty}^{p_0 + i\infty} e^{pt}\tilde{f}(p)\,dp$$

pic 01 - invs lap integral



We want to transform $f_{\alpha 1}$ and ϕ_1

$$\text{Fourier transform}\begin{cases} f_{\alpha\bar{k}}(\bar{v},t) = \dfrac{1}{(2\pi)^3}\displaystyle\int_{-\infty}^{\infty} e^{-i\bar{k}\cdot\bar{r}} f_{\alpha 1}\left(\bar{r},\bar{v},t\right) d^3r \\ \phi_{\bar{k}}(t) = \dfrac{1}{(2\pi)^3}\displaystyle\int_{-\infty}^{\infty} e^{-i\bar{k}\cdot\bar{r}} \phi_1\left(t\right) d^3r \end{cases}$$

$$\text{Laplace transform}\begin{cases} \tilde{f}_{\alpha k}(\bar{v},p) = \displaystyle\int_0^{\infty} e^{-pt} f_{\alpha\bar{k}}(\bar{v},t)\, dt \\ \tilde{\phi}_k(p) = \displaystyle\int_0^{\infty} e^{-pt} \phi_{\bar{k}}(t)\, dt \end{cases}$$

for $\mathrm{Re}(p) \geq p_0$
and

$$f_{\alpha 1}(\bar{r},\bar{v},t) = \frac{1}{2\pi i}\int_{p_0-i\infty}^{p_0+i\infty} dp\, e^{pt} \int_{-\infty}^{\infty} e^{i\bar{k}\cdot\bar{r}} \tilde{f}_{\alpha\bar{k}}(\bar{v},p)\, d^3k$$

$$\phi_1(\bar{r},t) = \frac{1}{2\pi i}\int_{p_0-i\infty}^{p_0+i\infty} dp\, e^{pt} \int_{-\infty}^{\infty} e^{i\bar{k}\cdot\bar{r}} \tilde{\phi}_{\bar{k}}(p)\, d^3k$$

Vlasov equation in Laplace transform

We transform Vlasov equation in space using Fourier transform and in time using Laplace transform

$$\int_0^{\infty} e^{-pt}\, dt \int_{-\infty}^{\infty} \frac{e^{-i\bar{k}\cdot\bar{r}}}{(2\pi)^3} d^3r \left[\frac{\partial f_{\alpha 1}}{\partial t} + \bar{v}\cdot\nabla f_{\alpha 1} - \frac{\rho_\alpha}{m_a}\nabla\phi_1\cdot\nabla_{\bar{v}} f_{\alpha 0}\right] = 0$$

$$\int_0^{\infty} e^{-pt}\, dt \left[\frac{\partial}{\partial t}\int_{-\infty}^{\infty} \frac{e^{-i\bar{k}\cdot\bar{r}}}{(2\pi)^3} f_{\alpha 1}\, d^3r + \bar{v}\int_{-\infty}^{\infty} \frac{e^{-i\bar{k}\cdot\bar{r}}}{(2\pi)^3}\nabla f_{\alpha 1}\, d^3r - \frac{\rho_\alpha}{m_a}\nabla_{\bar{v}} f_{\alpha 0}\cdot\int_{-\infty}^{\infty}\frac{e^{-i\bar{k}\cdot\bar{r}}}{(2\pi)^3}\nabla\phi_1\, d^3r\right] = 0$$

The **first integral** becomes

$$I_1 = \int_{-\infty}^{\infty} \frac{e^{-i\bar{k}\cdot\bar{r}}}{(2\pi)^3} f_{\alpha 1}\, d^3r = f_{\alpha\bar{k}}(\bar{v},t)$$

The **second integral** becomes

$$I_2 = \int_{-\infty}^{\infty} \frac{e^{-i\bar{k}\cdot\bar{r}}}{(2\pi)^3} \nabla f_{\alpha 1}\, d^3r = \frac{1}{(2\pi)^3}\left(\int_{-\infty}^{\infty} e^{-i\bar{k}\cdot\bar{r}} \frac{\partial f_{\alpha 1}}{\partial x}\, d^3r,\ \int_{-\infty}^{\infty} e^{-i\bar{k}\cdot\bar{r}} \frac{\partial f_{\alpha 1}}{\partial y}\, d^3r,\ \int_{-\infty}^{\infty} e^{-i\bar{k}\cdot\bar{r}} \frac{\partial f_{\alpha 1}}{\partial z}\, d^3r\right)$$

and the integral in the x direction is (solving by parts)

$$\begin{aligned}
\int_{-\infty}^{\infty} e^{-i\bar{k}\cdot\bar{r}} \frac{\partial f_{\alpha 1}}{\partial x}\, d^3r &= \int_{-\infty}^{\infty} dz\, e^{-ik_z z} \int_{-\infty}^{\infty} dy\, e^{-ik_y y} \int_{-\infty}^{\infty} dx\, e^{-ik_x x} \left(\frac{\partial f_{\alpha 1}}{\partial x}\right) \\
&= \int_{-\infty}^{\infty} dz\, e^{-ik_z z} \int_{-\infty}^{\infty} dy\, e^{-ik_y y} \left[\cancel{f_{\alpha 1} e^{-ik_x x}\Big|_{-\infty}^{+\infty}} - \int_{-\infty}^{\infty} f_{\alpha 1}\, (-ik_x) e^{-ik_x x}\, dx\right] \\
&= \int_{-\infty}^{\infty} dz\, e^{-ik_z z} \int_{-\infty}^{\infty} dy\, e^{-ik_y y} \int_{-\infty}^{\infty} (ik_x) f_{\alpha 1} e^{-ik_x x}\, dx \\
&= (ik_x) \int_{-\infty}^{\infty} dz\, e^{-ik_z z} \int_{-\infty}^{\infty} dy\, e^{-ik_y y} \int_{-\infty}^{\infty} f_{\alpha 1} e^{-ik_x x}\, dx \\
&= (ik_x) \int_{-\infty}^{\infty} f_{\alpha 1} e^{-i\bar{k}\cdot\bar{r}}\, d^3r
\end{aligned}$$

we can cancel that term because $f_{\alpha 1}$ goes to zero at infinity
for y and z we have

$$\begin{aligned}
y: &\qquad ik_y \int_{-\infty}^{\infty} f_{\alpha 1} e^{-i\bar{k}\cdot\bar{r}}\, d^3r \\
z: &\qquad ik_z \int_{-\infty}^{\infty} f_{\alpha 1} e^{-i\bar{k}\cdot\bar{r}}\, d^3r
\end{aligned}$$

so

$$I_2 = i\bar{k}\, f_{\alpha\bar{k}}(\bar{v}, t)$$

The **third integral** becomes (using the same reasoning as before)

$$I_3 = \int_{-\infty}^{\infty} \frac{e^{-i\bar{k}\cdot\bar{r}}}{(2\pi)^3} \nabla\phi_1\, d^3r = i\bar{k}\, \phi_{\bar{k}}(t)$$

So we have

$$\int_0^{\infty} e^{-pt}\, dt \left[\frac{\partial}{\partial t} f_{\alpha\bar{k}}(\bar{v}, t) + i\bar{v}\cdot\bar{k}\, f_{\alpha\bar{k}}(\bar{v}, t) - \frac{\rho_\alpha}{m_a} \nabla_{\bar{v}} f_{\alpha 0} \cdot i\bar{k}\, \phi_{\bar{k}}(t)\right] = 0$$

Now we solve the time integral
The **first integral** becomes

$$\begin{aligned}
\int_0^{\infty} e^{-pt} \frac{\partial}{\partial t} f_{\alpha\bar{k}}(\bar{v}, t)\, dt &= e^{-pt} f_{\alpha\bar{k}}(\bar{v}, t)\big|_0^{\infty} - \int_0^{\infty} \left[-pe^{-pt} f_{\alpha\bar{k}}(\bar{v}, t)\right] dt \\
&= -f_{\alpha\bar{k}}(\bar{v}, t=0) + p\tilde{f}_{\alpha\bar{k}}(\bar{v}, p)
\end{aligned}$$

The **second integral** becomes

$$\int_0^{\infty} e^{-pt} \left(i\bar{v}\cdot\bar{k}\right) f_{\alpha\bar{k}}(\bar{v}, t)\, dt = i\bar{v}\cdot\bar{k}\, \tilde{f}_{\alpha\bar{k}}(\bar{v}, p)$$

The **third integral** becomes

$$\int_0^\infty e^{-pt}\left(\frac{\rho_\alpha}{m_a}\nabla_{\bar{v}} f_{\alpha 0}\cdot i\bar{k}\right)\phi_{\bar{k}}(t)\,dt = \frac{\rho_\alpha}{m_a}\nabla_{\bar{v}} f_{\alpha 0}\cdot i\bar{k}\,\tilde{\phi}_{\bar{k}}(p)$$

And the full equation becomes

$$-f_{\alpha\bar{k}}(\bar{v}, t=0) + p\tilde{f}_{\alpha\bar{k}}(\bar{v}, p) + i\bar{v}\cdot\bar{k}\;\tilde{f}_{\alpha\bar{k}}(\bar{v}, p)\; - i\frac{\rho_\alpha}{m_a}\nabla_{\bar{v}} f_{\alpha 0}\cdot\bar{k}\,\tilde{\phi}_{\bar{k}}(p) = 0$$

Poisson equation in Laplace transform

$$-\nabla^2\phi_1 = 4\pi\sum_\alpha \rho_\alpha\int_{-\infty}^{\infty} f_{\alpha 1}\left(\bar{r}, \bar{v}, t\right)d^3v$$

We apply the transformations as before

$$\int_0^\infty e^{-pt}\,dt\int_{-\infty}^{\infty}\frac{e^{-i\bar{k}\cdot\bar{r}}}{(2\pi)^3}\left[\nabla^2\phi_1 + 4\pi\sum_\alpha\rho_\alpha\int_{-\infty}^{\infty} f_{\alpha 1}\,d^3v\right]d^3r = 0$$

The **second term** becomes

$$\int_0^\infty e^{-pt}\,dt\int_{-\infty}^{\infty}\frac{e^{-i\bar{k}\cdot\bar{r}}}{(2\pi)^3}\left[4\pi\sum_\alpha\rho_\alpha\int_{-\infty}^{\infty} f_{\alpha 1}\,d^3v\right]d^3r =$$

$$= 4\pi\sum_\alpha\rho_\alpha\int_{-\infty}^{\infty}\left[\int_0^\infty e^{-pt}\,dt\int_{-\infty}^{\infty}\frac{e^{-i\bar{k}\cdot\bar{r}}}{(2\pi)^3}f_{\alpha 1}\,d^3r\right]d^3v =$$

$$= 4\pi\sum_\alpha\rho_\alpha\int_{-\infty}^{\infty}\tilde{f}_{\alpha\bar{k}}\,d^3v$$

Now we consider the Fourier transformation of the **first term**, which becomes

$$\int_{-\infty}^{\infty} e^{-i\bar{k}\cdot\bar{r}}\,\nabla^2\phi_1\,d^3r = \int_{-\infty}^{\infty} e^{-i\bar{k}\cdot\bar{r}}\,\left(\frac{\partial^2\phi_1}{\partial x^2} + \frac{\partial^2\phi_1}{\partial y^2} + \frac{\partial^2\phi_1}{\partial z^2}\right)d^3r =$$

$$= \int_{-\infty}^{\infty} dz\,e^{-ik_z z}\int_{-\infty}^{\infty} dy\,e^{-ik_y y}\int_{-\infty}^{\infty} dx\,e^{-ik_x x}\left(\frac{\partial^2\phi_1}{\partial x^2}\right)+$$

$$+\int_{-\infty}^{\infty} dz\,e^{-ik_z z}\int_{-\infty}^{\infty} dx\,e^{-ik_x x}\int_{-\infty}^{\infty} dy\,e^{-ik_y y}\left(\frac{\partial^2\phi_1}{\partial y^2}\right)+$$

$$+\int_{-\infty}^{\infty} dx\,e^{-ik_x x}\int_{-\infty}^{\infty} dy\,e^{-ik_y y}\int_{-\infty}^{\infty} dz\,e^{-ik_z z}\left(\frac{\partial^2\phi_1}{\partial z^2}\right)$$

Now, we solve the integral by parts two times and we use the fact that the electric potential (and also its derivative) goes to zero at infinity. We have

$$\int_{-\infty}^{\infty} e^{-ik_x x} \frac{\partial^2 \phi_1}{\partial x^2}\, dx = \cancel{\frac{\partial \phi_1}{\partial x} e^{-ik_x x}\Big|_{-\infty}^{+\infty}} - \int_{-\infty}^{+\infty} (-ik_x) \frac{\partial \phi_1}{\partial x} e^{-ik_x x}\, dx =$$

$$= ik_x \int_{-\infty}^{+\infty} \frac{\partial \phi_1}{\partial x} e^{-ik_x x}\, dx =$$

$$= \cancel{ik_x \phi_1 e^{-ik_x x}\Big|_{-\infty}^{+\infty}} - ik_x \int_{-\infty}^{+\infty} (-ik_x) \phi_1 e^{-ik_x x}\, dx =$$

$$= -k_x^2 \int_{-\infty}^{+\infty} \phi_1 e^{-ik_x x}\, dx$$

And the same reasoning works for the other two integrals in y and z
So we have

$$\int_{-\infty}^{\infty} \frac{e^{-i\bar{k}\cdot\bar{r}}}{(2\pi)^3} \nabla^2 \phi_1\, d^3r = -k_x^2 \int_{-\infty}^{\infty} \frac{e^{-i\bar{k}\cdot\bar{r}}}{(2\pi)^3} \phi_1\, d^3r - k_y^2 \int_{-\infty}^{\infty} \frac{e^{-i\bar{k}\cdot\bar{r}}}{(2\pi)^3} \phi_1\, d^3r - k_z^2 \int_{-\infty}^{\infty} \frac{e^{-i\bar{k}\cdot\bar{r}}}{(2\pi)^3} \phi_1\, d^3r =$$

$$= -\left(k_x^2 + k_y^2 + k_z^2\right) \cdot \int_{-\infty}^{\infty} \frac{e^{-i\bar{k}\cdot\bar{r}}}{(2\pi)^3} \phi_1\, d^3r =$$

$$= -k^2\, \phi_{\bar{k}}(t)$$

Now we substitute these results and we get

$$\int_0^{\infty} e^{-pt}\, dt \left(-k^2\, \phi_{\bar{k}}(t)\right) + 4\pi \sum_\alpha \rho_\alpha \int_{-\infty}^{\infty} \tilde{f}_{\alpha\bar{k}}\, d^3v = 0$$

The first term actually is

$$-k^2\, \tilde{\phi}_{\bar{k}}(p) + 4\pi \sum_\alpha \rho_\alpha \int_{-\infty}^{\infty} \tilde{f}_{\alpha\bar{k}}\, d^3v = 0$$

and finally we have obtained the

Vlasov-Poisson equations in Laplace-Fourier space

$$\begin{cases} -f_{\alpha\bar{k}}(\bar{v}, t=0) + p\, \tilde{f}_{\alpha\bar{k}}(\bar{v}, p) + i\bar{v}\cdot\bar{k}\, \tilde{f}_{\alpha\bar{k}}(\bar{v}, p)\ - i\dfrac{\rho_\alpha}{m_a} \bar{k}\cdot\nabla_{\bar{v}} f_{\alpha 0}\, \tilde{\phi}_{\bar{k}}(p) = 0 \\ -k^2\, \tilde{\phi}_{\bar{k}}(p) + 4\pi \displaystyle\sum_\alpha \rho_\alpha \int_{-\infty}^{\infty} \tilde{f}_{\alpha\bar{k}}\, d^3v = 0 \end{cases}$$

In particular, we can isolate the electric potential.
First we isolate the distribution function

$$\tilde{f}_{\alpha\bar{k}}(\bar{v}, p) \left[p + i\bar{v}\cdot\bar{k}\right] = f_{\alpha\bar{k}}(\bar{v}, t=0) + i\frac{\rho_\alpha}{m_a} \bar{k}\cdot\nabla_{\bar{v}} f_{\alpha 0}\, \tilde{\phi}_{\bar{k}}(p)$$

$$\Longrightarrow \tilde{f}_{\alpha\bar{k}}(\bar{v}, p) = \frac{f_{\alpha\bar{k}}(\bar{v}, t=0) + i\dfrac{\rho_\alpha}{m_a} \bar{k}\cdot\nabla_{\bar{v}} f_{\alpha 0}\, \tilde{\phi}_{\bar{k}}(p)}{p + i\bar{v}\cdot\bar{k}}$$

Then we substitute it in the Poisson equation

$$k^2\,\tilde{\phi}_{\bar{k}}(p) = 4\pi \sum_\alpha \rho_\alpha \int_{-\infty}^{\infty} \left[\frac{f_{\alpha\bar{k}}(\bar{v}, t=0) + i\dfrac{\rho_\alpha}{m_a}\bar{k}\cdot\nabla_{\bar{v}} f_{\alpha 0}\,\tilde{\phi}_{\bar{k}}(p)}{p + i\bar{v}\cdot\bar{k}} \right] d^3v$$

$$\implies k^2\,\tilde{\phi}_{\bar{k}}(p) = 4\pi \sum_\alpha \rho_\alpha \int_{-\infty}^{\infty} \frac{f_{\alpha\bar{k}}(\bar{v}, t=0)}{p + i\bar{v}\cdot\bar{k}}\, d^3v + 4\pi i \sum_\alpha \frac{\rho_\alpha^2}{m_a}\bar{k}\,\tilde{\phi}_{\bar{k}}(p)\cdot \int_{-\infty}^{\infty} \frac{\nabla_{\bar{v}} f_{\alpha 0}}{p + i\bar{v}\cdot\bar{k}}\, d^3v$$

$$\implies k^2\,\tilde{\phi}_{\bar{k}}(p) \left[1 - \frac{4\pi}{k^2} i \sum_\alpha \frac{\rho_\alpha^2}{m_a}\bar{k}\cdot \int_{-\infty}^{\infty} \frac{\nabla_{\bar{v}} f_{\alpha 0}}{p + i\bar{v}\cdot\bar{k}}\, d^3v\right] = 4\pi \sum_\alpha \rho_\alpha \int_{-\infty}^{\infty} \frac{f_{\alpha\bar{k}}(\bar{v}, t=0)}{p + i\bar{v}\cdot\bar{k}}\, d^3v$$

$$\implies k^2\,\tilde{\phi}_{\bar{k}}(p) = \frac{4\pi \displaystyle\sum_\alpha \rho_\alpha \int_{-\infty}^{\infty} \frac{f_{\alpha\bar{k}}(\bar{v}, t=0)}{p + i\bar{v}\cdot\bar{k}}\, d^3v}{1 - \dfrac{4\pi}{k^2} i \displaystyle\sum_\alpha \frac{\rho_\alpha^2}{m_a}\bar{k}\cdot \int_{-\infty}^{\infty} \frac{\nabla_{\bar{v}} f_{\alpha 0}}{p + i\bar{v}\cdot\bar{k}}\, d^3v}$$

Previous lecture: [2025-10-10 - Valentini (4)]
Next lecture: [2025-10-17 - Valentini (6)]